Nutanix Certified Associate (NCA) 6.10

Manual PRO Study Guide - Exam-Focused Edition

Neutral study guide based on the official NCA 6.10 blueprint and the provided NHCF study modules. This guide is for learning and review only; the practice question bank is delivered separately for simulator use.

How to Use This Guide

- Study by objective, not by memorizing isolated terms.
- Associate every concept with where it appears in Prism Central or Prism Element.
- Use the separate question bank for simulator practice; do not mix quiz questions into the study guide.
- Prioritize LCM, VM operations, health checks, alerts/events, storage resiliency, AHV networking, licensing, and DR basics.
- Treat third-party question dumps as topic clues only; validate answers against blueprint and study material.

Exam Scope Snapshot

Item	NCA 6.10 Focus
Primary skill	Navigate Prism UI, interpret UI information, and perform basic operational tasks.
Exam format	Multiple choice and multiple response.
Question count	50 questions.
Time limit	90 minutes.
Core software versions	AOS 6.10 and Prism Central pc2024.2.
Recommended background	General IT infrastructure experience plus Nutanix virtualization exposure or equivalent training.

Objective-to-Module Map

NCA Objective Area	High-value Topics	Primary Study Module
Section 1 - Lifecycle Management	LCM inventory, upgrade order, prechecks, task status, maintenance mode	Module 8 - Licensing and Upgrades
Section 2 - Basic Administration	VM snapshots, migration, resource monitoring, affinity, cluster settings, licensing	Modules 1, 4, 5, 8
Section 3 - Environmental Health	NCC, resiliency, alerts/events, logs, Pulse, support cases	Module 7 plus Module 1
Section 4 - Cluster Configuration	RF/RF3, storage components, optimization, hypervisors, AHV networking, DR products	Modules 2, 3, 6

1. Lifecycle Management (LCM)

Core idea

LCM is the Nutanix lifecycle framework used to inventory software/firmware versions, discover available updates, validate dependencies, run prechecks, and execute upgrades in a controlled way.

What to know

- LCM Inventory identifies installed versions and available updates.
- Resolve critical alerts and run health checks before upgrades.
- LCM exists in Prism Element and Prism Central, but scope differs.
- Dark sites use local web server or direct upload workflows.
- LCM updates are planned, node-by-node, and not treated as simple rollback operations.
- Maintenance mode migrates eligible VMs, marks the host unschedulable, and powers down services as required.

Prism focus

- Prism Element: cluster components such as AOS, AHV, NCC, File Server and firmware.
- Prism Central: PC-local services and centralized lifecycle views.
- Tasks/Task Status: confirm upgrade and inventory progress.

High-probability checks

- What does Perform Inventory do?
- What is Direct Upload used for?
- Why run NCC/prechecks before upgrades?
- What happens when a node enters maintenance mode?
- Why should upgrades wait for resiliency to return to normal?

2. Nutanix Basic Administration

Core idea

NCA expects basic operational ability in Prism: create, inspect, update, migrate, snapshot, and monitor VMs; identify cluster settings; and understand licensing status.

VM tasks

- Create and manage VMs from Prism Central or Prism Element.
- Use images/ISOs from Image Service for operating system installation.
- Use VirtIO drivers for Windows VMs on AHV.
- Create snapshots in Prism Element and Recovery Points in Prism Central.
- Migrate VMs to another host for maintenance or resource balancing.
- Use categories and affinity when placement policies are needed.

Cluster-level settings

- RBAC controls what users can do.
- NTP keeps logs, tasks, snapshots, authentication, and DR schedules aligned.
- SMTP enables alert email notifications.
- Data-at-Rest Encryption protects stored data.
- Recycle Bin provides short-term protection for deleted vDisk/configuration files when capacity allows.

Licensing

- License Manager provides licensing services inside the cluster software.
- Seamless Licensing is preferred when Internet connectivity exists.
- Manual and dark-site licensing use file-based workflows.
- The Support Portal shows purchased inventory; Prism shows applied cluster licensing.

3. Cluster Health, Monitoring, Alerts and Support

Core idea

The environmental health section focuses on reading cluster condition and collecting evidence: health checks, NCC, resiliency, alerts, events, audits, Pulse, logs, and support cases.

Health workflow

- Start with Health and Data Resiliency Status.
- Run NCC before upgrades or when investigating issues.
- Collect logs from Prism Element when support evidence is needed.
- Use Analysis for metric correlation and Reports for historical reporting.

Monitoring distinctions

- Metric Chart: one metric across one or more entities.
- Entity Chart: one entity with one or more metrics.
- Alerts may require acknowledgment/resolution.
- Events are informational records.
- Audits provide traceability of user/system actions.

Pulse and support

- Pulse sends diagnostic data to Nutanix over HTTPS.
- Data obfuscation helps protect customer-private information.
- Insights and Discoveries can identify known issues or risks.
- Support cases should include symptoms, affected entity, timestamps, alerts/events, and log bundles.

4. Storage, Resiliency and Optimization

Core idea

Nutanix storage is distributed. The exam expects you to understand containers, storage pools, RF, resiliency, volume groups, data locality, tiering, and optimization features.

Key hierarchy

- Node and disks provide physical resources.
- AOS Storage aggregates resources into a distributed storage platform.
- Storage pool aggregates physical storage resources.
- Storage container stores vDisks and applies policies like RF, compression, deduplication, and erasure coding.
- Volume Group groups related vDisks and can support block/iSCSI use cases.

Resiliency

- Replication Factor = number of data/metadata copies.
- Redundancy Factor = cluster failure tolerance.
- RF2 keeps two data copies; RF3 adds another copy and requires more nodes/failure domains.
- RF1 should only be used when data is disposable or the application provides its own resiliency.
- Data Resiliency Status indicates whether the cluster can sustain hardware failure.

Optimization

- Compression reduces capacity usage for compressible data.
- Deduplication reduces duplicate blocks; value depends on workload.
- Erasure Coding saves capacity but is not ideal for write-heavy/overwrite-heavy workloads.

- Data locality keeps VM data close to the VM for performance.
- Intelligent tiering moves hot/cold data across storage tiers.

5. AHV Networking

Core idea

AHV networking uses Open vSwitch, bridges, bonds, VLANs, subnets, and IPAM to connect VMs, CVMs, hosts, and physical networks.

Terms to master

- OVS: virtual switch used by AHV.
- Bridge: virtual switching structure, traditionally br0.
- Tap port: connects a VM vNIC to OVS.
- Bond: groups physical NICs for resiliency and/or load distribution.
- VLAN: logical network segmentation.
- Managed network: IPAM enabled.
- Unmanaged network: IP assigned manually or by external DHCP.

Bond modes

- Active-backup: default/simple failover behavior.
- Balance-SLB: distributes VM traffic across uplinks without LACP.
- Balance-TCP: uses LACP and requires proper upstream switch configuration.

Prism focus

- Prism Central: Subnets and Network & Security views.
- Prism Element: Network Visualizer for graphical cluster topology.

6. Data Protection and Disaster Recovery

Core idea

DR questions usually test RPO/RTO, snapshots/recovery points, protection domains, remote sites, availability zones, recovery plans, NearSync, Async DR, and Metro Availability.

Concepts

- RPO: maximum acceptable data loss.
- RTO: maximum acceptable recovery time.
- Snapshot: point-in-time capture; in Prism Central these appear as Recovery Points.
- Protection Domain: Prism Element grouping for protected entities and schedules.
- Protection Policy: Prism Central DR policy for workload protection.
- Recovery Plan: orchestrates recovery order, delays, and network mappings.

Replication timing

- NearSync: 1 to 15 minute RPO using lightweight snapshots.
- Asynchronous DR: 60 minutes or greater RPO using asynchronous replication.
- Metro Availability: synchronous replication for supported site/container configurations.

Exam caution

- Do not confuse Objects, Files, and Volumes. SQL/shared block use cases usually align with Volumes; S3 object storage aligns with Objects.

7. Nutanix Products and Services

Core idea

NCA includes basic product/service use cases, not deep design mastery.

High-value products

- AOS Storage: distributed storage platform.
- AHV: Nutanix native hypervisor.
- Prism Element: one-cluster management.
- Prism Central: centralized management and advanced services.
- Nutanix Volumes: block/iSCSI-style storage services.
- Nutanix Files: file services.
- Nutanix Objects: S3-compatible object storage.
- Nutanix Move: workload migration.
- NC2: Nutanix clusters in supported public clouds.
- Flow Virtual Networking: advanced virtual networking constructs.

Final Review Path

1. Review the objective-to-module map and identify weak sections.
2. Study Section 1 LCM until you can explain inventory, direct upload, upgrade order, and maintenance mode.
3. Practice VM basics: snapshot, recovery point, migration, categories, affinity, and templates.
4. Review Health: NCC, Data Resiliency Status, alerts/events/audits, logs, Pulse and support cases.
5. Master storage definitions: storage pool, container, volume group, RF2/RF3/RF1, compression, dedupe, erasure coding.
6. Review AHV networking: OVS, bonds, managed vs unmanaged networks, VLANs and Network Visualizer.
7. Review DR: RPO/RTO, Protection Domains, Protection Policies, Recovery Plans, NearSync, Async DR, Metro Availability.
8. Use the simulator question bank in short sessions, then review incorrect answers by objective.

Branch Recommendation for GitHub

Recommended branch name: nca-6.10-simulator

Keep this work separate from the existing NCP-MCI simulator branch. Store the question bank under a dedicated NCA folder, for example: data/nca610_question_bank.js or data/nca610_question_bank.json.

Question Bank Note

The simulator bank is delivered separately. Questions marked NEW_PRACTICE_QUESTION were created to improve coverage of official objectives. Questions marked CORRECTED_SOURCE_DERIVED were adapted from provided material because the provided answer or wording appeared questionable.